MATH 350-2 Advanced Calculus

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Outline

1. Real Analysis Lecture 20
   - More on differentiation
   - Riemann-Stieltjes integrals

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Local maxima and minima

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It has **local maximum at** $a \in S$ if there exists an $r > 0$ such that

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Absolute and local minima are defined similarly.

Critical points

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Theorem

Let f be a real-valued function on an open interval (a, b) and suppose that f has a local maximum or local minimum at $c \in (a, b)$. Then c is a critical point of f.

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If f is not differentiable at c, then we're done!
Therefore assume that it is differentiable.
By definition, there exists $r > 0$ such that

$$f(x) \leq f(c) \quad \text{for all } x \in (c - r, c + r) \cap (a, b).$$

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To obtain a contradiction, suppose that $f'(c) \neq 0$.

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Therefore assume that it is differentiable.
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$$f(x) \leq f(c) \quad \text{for all } x \in (c - r, c + r) \cap (a, b).$$

To obtain a contradiction, suppose that $f'(c) \neq 0$.
Then by the Helper Theorem, there is a function $f^*(x)$ defined on (a, b) and continuous at $c \in (a, b)$ with

$$f(x) - f(c) = (x - c)f^*(x).$$



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Since f is continuous, we may choose $\delta > 0$ such that

$$|x - c| < \delta \quad \Rightarrow \quad |f^*(x) - f^*(c)| < |f^*(c)|/2.$$

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Moreover, we may assume that $\delta < r$.
It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.

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It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.
However,

$$f(c \pm \delta/2) = f(c) \pm \frac{1}{2}\delta f^*(c + \delta/2),$$

one of these will be larger than $f(c)$.

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However,

$$f(c \pm \delta/2) = f(c) \pm \frac{1}{2}\delta f^*(c + \delta/2),$$

one of these will be larger than $f(c)$. Contradiction! □

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Then there exists $c \in (a, b)$ with $f'(c) = 0$.

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Then $f'(c) = 0$ for every $c \in (a, b)$.
Otherwise at least one of the points happens inside, so $f(x)$ has a local max or local min at a value c inside (a, b).

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If both of these occur at the endpoints, then the absolute max is the absolute min, forcing f to be constant.
Then $f'(c) = 0$ for every $c \in (a, b)$.
Otherwise at least one of the points happens inside, so $f(x)$ has a local max or local min at a value c inside (a, b).
The point c is a critical point, and since f is differentiable at c, we must have $f'(c) = 0$. □

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Then there exists $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Mean Value Theorem

Proof.

Consider the function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a).$$

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Then $g(x)$ is

- continuous on $[a, b]$;
- differentiable at each point in (a, b);
- and satisfies $g(a) = g(b)$.

By Rolle's Theorem, there exists $c \in (a, b)$ with

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$



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Most of these generalize the Riemann integral in some sense.

Partitions

Definition

A **partition** P of $[a, b]$ Is a set of points

$$\{x_0, x_1, \ldots, x_n\}$$

with

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

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A set of **sample points** for P is a set $\{t_1, \ldots, t_n\}$ with $t_k \in [x_{k-1}, x_k]$ for all $1 \leq k \leq n$. A partition P' is a **refinement** of P if $P \subseteq P'$.

Riemann-Stieltjes sums

Notation: given a real-valued function α on $[a, b]$

$$\Delta\alpha_k(P) = \alpha(x_k) - \alpha(x_{k-1}), \quad k = 1, 2, \ldots, n.$$

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Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$.

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Definition

Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. A sum of the form

$$S(P, f, \alpha, \{t_k\}) = \sum_{k=1}^{n} f(t_k)\Delta\alpha_k(P),$$

for some set of sample points $\{t_k\}$ of P.

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Let f and α be real-valued functions on $[a, b]$. We say f is **Riemann integrable** with respect to α on $[a, b]$ if there exists $A \in \mathbb{R}$ with the following property: for all $\epsilon > 0$, there exists a partition P_ϵ of $[a, b]$ such that for all refinements P of P_ϵ

$$|S(P, f, \alpha, \{t_k\}) - A| < \epsilon,$$

for all sets of sample points $\{t_k\}$ of P.

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for all sets of sample points $\{t_k\}$ of P.

Notation: in this case A is denote by $\int_a^b f d\alpha$ or $\int_a^b f(x) d\alpha(x)$

Linearity of integration

Theorem

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Suppose that f and g are Riemann integrable with respect to α on $[a, b]$.
Then for any $c_1, c_2 \in \mathbb{R}$, $c_1 f + c_2 g$ is Riemann integrable with respect to α and

$$\int_a^b (c_1 f + c_2 g) d\alpha = c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha.$$

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Choose a partition P'_ϵ such that

$$|S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| < \epsilon/2|c_1|,$$

for all refinements P of P'_ϵ and sets of sample points $\{t_k\}$ of P.

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for all refinements P of P'_ϵ and sets of sample points $\{t_k\}$ of P.
Also, choose a partition P''_ϵ such that

$$|S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha| < \epsilon/2|c_2|,$$

for all refinements P of P''_ϵ and sets of sample points $\{t_k\}$ of P.

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Let $P_\epsilon = P'_\epsilon \cup P''_\epsilon$.

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for all refinements P of P'_ϵ and sets of sample points $\{t_k\}$ of P.
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for all refinements P of P''_ϵ and sets of sample points $\{t_k\}$ of P.
Let $P_\epsilon = P'_\epsilon \cup P''_\epsilon$. Then any refinement P of P_ϵ, is a refinement of P'_ϵ and P''_ϵ. □

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Therefore if P is a refinement of P_ϵ, and $\{t_k\}$ is a set of sample points of P, then by the triangle inequality

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Therefore if P is a refinement of P_ϵ, and $\{t_k\}$ is a set of sample points of P, then by the triangle inequality

$$|S(P, c_1 f + c_2 g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)|$$

$$= |c_1 S(P, f, \alpha, \{t_k\}) + c_2 S(P, g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)|$$

$$= |c_1 S(P, f, \alpha, \{t_k\}) - c_1 \int_a^b f d\alpha + c_2 S(P, g, \alpha, \{t_k\}) - c_2 \int_a^b g d\alpha|$$

$$\leq |c_1| \cdot |S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| + |c_2| \cdot |S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha|$$

$$< \epsilon/2 + \epsilon/2 = \epsilon.$$

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Proof.

Therefore if P is a refinement of P_ϵ, and $\{t_k\}$ is a set of sample points of P, then by the triangle inequality

$$|S(P, c_1 f + c_2 g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)|$$

$$= |c_1 S(P, f, \alpha, \{t_k\}) + c_2 S(P, g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)|$$

$$= |c_1 S(P, f, \alpha, \{t_k\}) - c_1 \int_a^b f d\alpha + c_2 S(P, g, \alpha, \{t_k\}) - c_2 \int_a^b g d\alpha|$$

$$\leq |c_1| \cdot |S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| + |c_2| \cdot |S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha|$$

$$< \epsilon/2 + \epsilon/2 = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this proves the theorem. □

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Suppose that f is Riemann integrable with respect to both α and β on $[a, b]$.
Then for any $c_1, c_2 \in \mathbb{R}$, f is Riemann integrable with respect to $c_1\alpha + c_2\beta$ and

$$\int_a^b f d(c_1\alpha + c_2\beta) = c_1 \int_a^b f d\alpha + c_2 \int_a^b f d\beta.$$

Challenge!

Problem

Prove the last theorem!

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Suppose that f is a real-valued function on $[a, b]$ and that $c \in (a, b)$.

Linearity of integration

Theorem

Suppose that f is a real-valued function on $[a, b]$ and that $c \in (a, b)$. If $\int_a^c f d\alpha$ and $\int_b^c f d\alpha$ exist, then $\int_a^c f d\alpha$ exists and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

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Suppose that f is a real-valued function on $[a, b]$ and that $c \in (a, b)$. If $\int_a^c f d\alpha$ and $\int_b^c f d\alpha$ exist, then $\int_a^c f d\alpha$ exists and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

Conversely, if $\int_a^c f d\alpha$ exists then $\int_a^c f d\alpha$ and $\int_b^c f d\alpha$ exist and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$